2. Central Nervous System - Structure and Function

General Objectives

- **Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system**

Content

The major divisions of the nervous system. The somatic and autonomic divisions of the peripheral nervous system, their roles and interrelationships, especially the relationship between the control centres of the medulla and the autonomic nervous system.
The CNS, parts of the brain – cerebrum, cerebellum, hypothalamus, medulla, meninges – and their functions; the spinal cord and cerebrospinal fluid.
Parts of the brain and nerve pathways (names of nerves not required) in relation to the voluntary control of locomotion and object manipulation, and involuntary control of posture and balance.

1. 1999 / 13

The part of the brain associated with thought and reasoning is the

(a) cerebellum.
(b) medulla oblongata.
(c) cerebrum.
(d) hypothalamus.

2. 1999 / 15

Normal cerebrospinal fluid

(a) contains red blood cells, proteins and fluids.
(b) causes the brain to change shape in response to a blow to the head.
(c) is secreted and absorbed so as to maintain a constant pressure.
(d) flows out from the spinal cord around the motor neurons.

3. 1999 / 29

The major control centres for respiration **and** heart rate are located in the

(a) pons.
(b) cerebellum.
(c) pituitary.
(d) medulla oblongata.

4. 2000 / 15

Which of this list is **NOT** a part of the body's peripheral nervous system?

(a) Cranial nerves
(b) Spinal nerves
(c) Spinal cord
(d) Touch receptor

5. 2000 / 16

Name the part of the brain that regulates body temperature, hunger, thirst, and sleep.

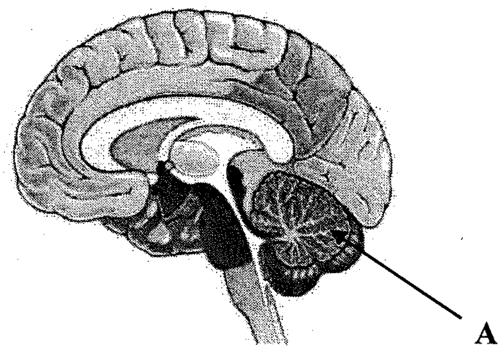
(a) Medulla
(b) Hypothalamus
(c) Cerebrum
(d) Pons

6. 2001 / 3
The spinal cord has white matter

(a) inside, grey matter outside.
(b) outside, grey matter inside and a ventral motor root.
(c) outside, grey matter inside and a dorsal motor root.
(d) inside, grey matter outside and a ventral motor root.

7. 2001 / 6
Damage to structure **A** in the figure that follows of the brain would result in

(a) uncoordinated movement.
(b) loss of memory.
(c) inability to regulate breathing.
(d) impaired hearing.



8. 2002 / 11
The cerebellum does **NOT** control

(a) posture.
(b) coordination.
(c) balance.
(d) body temperature.

9. 2003 / 12
Which of the following is an example of an effector in a reflex arc?

(a) Pain receptor in the foot
(b) Motor nerve fibres in the spinal cord
(c) Motor end plate
(d) Skeletal muscle

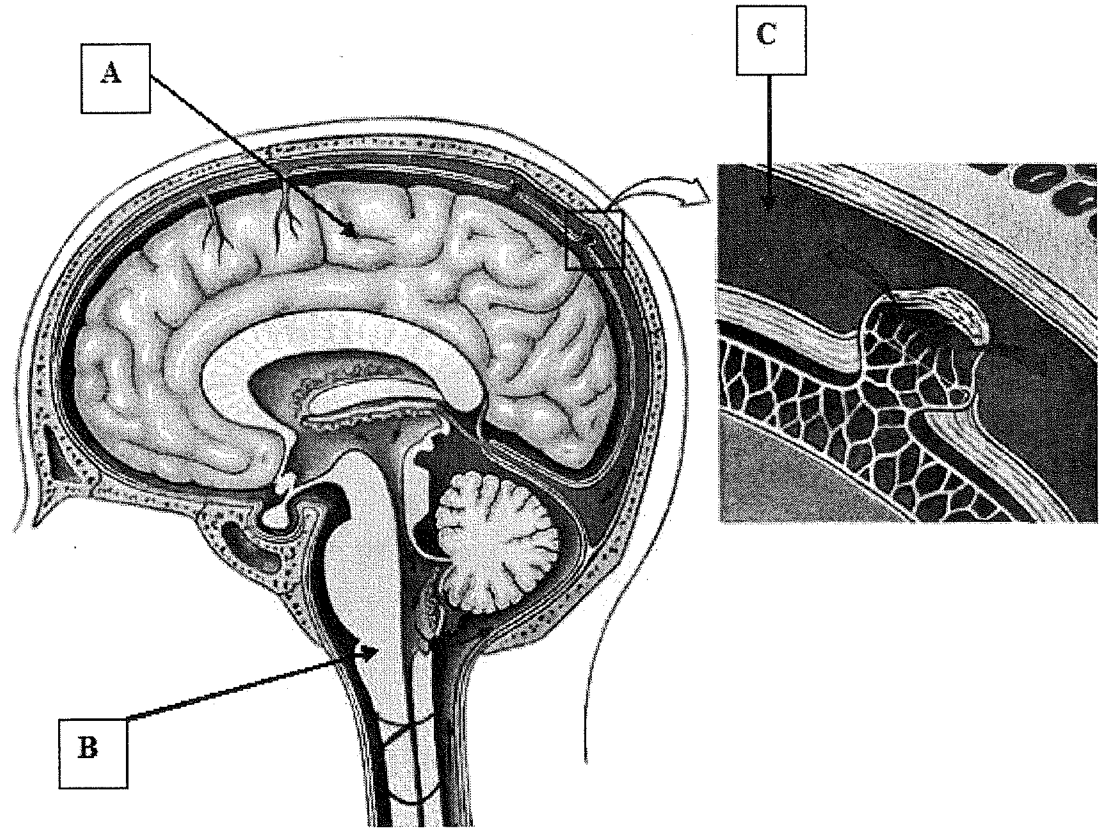
10. 2003 / 13
If the ventral root of a spinal cord is cut, what will be the result in the tissue or region that the nerve supplies?

(a) Complete loss of sensation
(b) Complete loss of voluntary movement
(c) Loss of neither sensation nor movement but only of autonomic control
(d) A complete loss of both sensation and movement

11. 2003 / 17
Injury to the hypothalamus may result in all of the following **EXCEPT**

(a) loss of awareness of body position.
(b) a loss of temperature control.
(c) a production of excessive quantities of urine.
(d) a disrupted menstrual cycle.

The next two questions refer to the diagram below.



12. 2004 / 1
Which of the following is **NOT** true of structure A?

(a) It is surrounded by three layers of meninges.
(b) Information from the body terminates in structure A's white matter.
(c) Its surface is convoluted to provide greater surface area.
(d) It is connected to, and able to influence, the cerebellum.

13. 2004 / 2
This question refers to the list of features below.

(i) regulation of osmotic balance
(ii) regulation of the heart rate
(iii) coordination of posture and movement
(iv) temperature control

Which of the above features are roles played by structure B?

(a) (i), (ii) and (iii) only
(b) (i), (ii) and (iv) only
(c) (i), (ii), (iii) and (iv) only
(d) (ii) and (iv) only

14. 2004 / 5
Which of the following statements is correct?

(a) The sensation of touch is processed in the cerebellum.
(b) The ventral root of a spinal nerve contains sensory neurons.
(c) The dorsal root ganglia contains the cell bodies of many motor neurons.
(d) Following the response, reflex actions are interpreted by the cerebrum.

15. 1999 / 49

A spinal reflex allows us to withdraw quickly from a painful stimulus. Use the diagram below of a foot standing on a nail, to help you name the structures that make up a spinal reflex arc. (5)

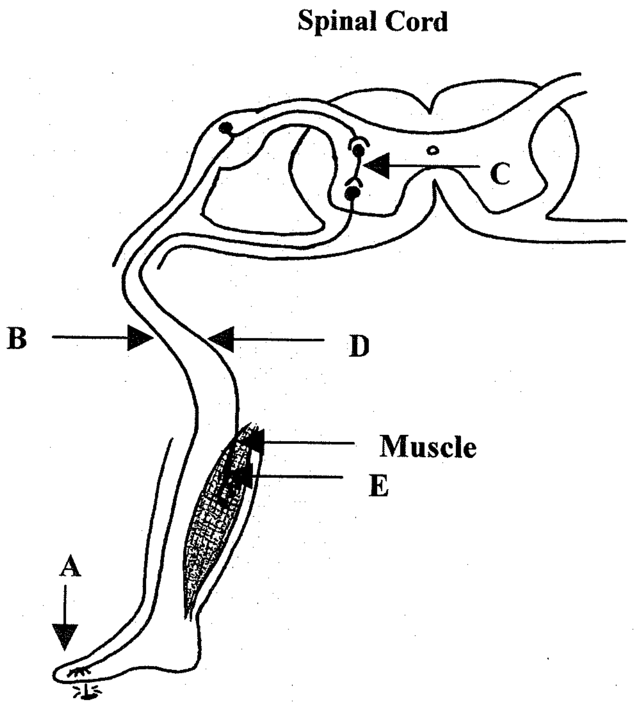
A ______________________

B ______________________

C ______________________

D ______________________

E ______________________



16. 2000 / 49

(c) Describe why death could result if a small area of the medulla oblongata is damaged, while similar damage to the cerebrum may not be fatal. (2)

____

____

____

____

(f) Cerebrospinal fluid is formed from the blood. It circulates throughout the Central Nervous System, and is gradually reabsorbed into the blood. Why does cerebrospinal fluid circulate in this manner? (2)

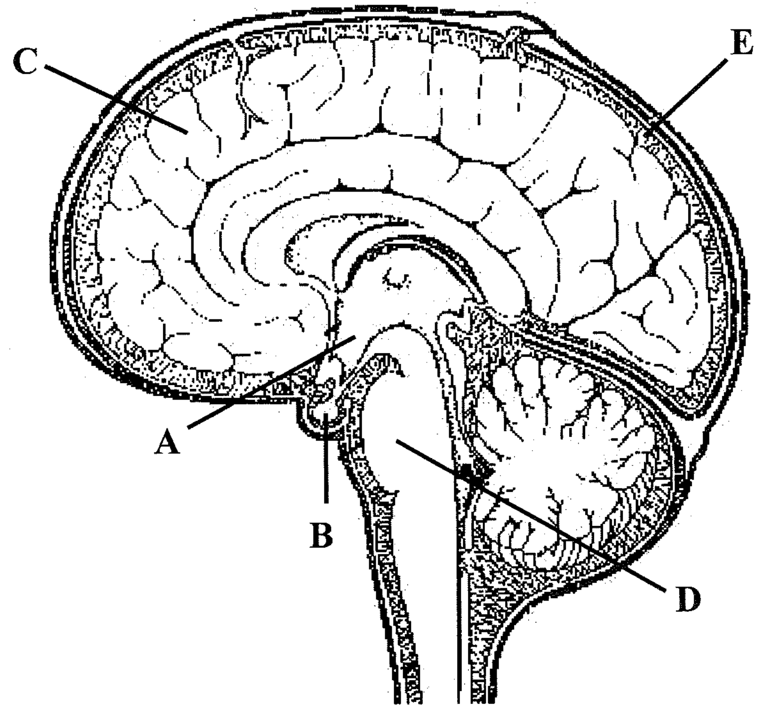
____

____

____

____

17. 2002 / 42



(a) Identify the labelled structures

A ______________________ (1)

B ______________________ (1)

C ______________________ (1)

D ______________________ (1)

(b) What is contained in the space labelled E?

______________________ (1)

EXTENDED ANSWERS

18. 2002 / 49

(b) An individual's spinal cord is severed below the level of the shoulder in a car accident. When examined, it is noticed that the patient's leg moves in response to a sharp stimulus to the foot. Explain how this response can occur. Include in your answer details of specific components of the pathway involved in this response. (7)

19. 2003 / 49

(b) Hospital patients who are in a coma experience a loss of function in many parts of their cerebrum and yet their heart continues to function normally.

Give an explanation for how the body is able to achieve this in the absence of a fully functioning cerebrum. (7)

3. Nervous System - Receptor Organs

<u>General Objectives</u>

- **Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system**

<u>Content</u>

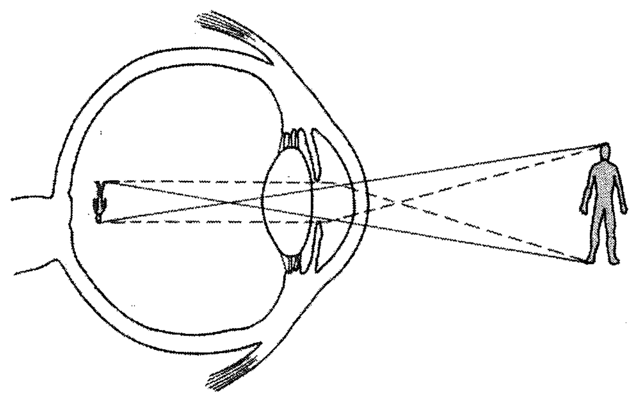
The existence of receptors for external and internal stimuli, and the role of the senses such as hearing, balance and vision. Structure, function and defects of the eye and ear.

THE EYE

1. 1999 / 14
One function of the cornea of the eye is to

(a) convert light into nervous impulses.
(b) allow light to enter the eye.
(c) regulate the amount of light entering the eye.
(d) maintain the shape of the eyeball.

2. 2002 / 9
Examine the diagram below that shows light rays entering an eye. What condition is illustrated in this diagram and how is this corrected?



(a) Myopia, corrected with a concave lens.
(b) Myopia, corrected with a convex lens.
(c) Hypermetropia, corrected with a concave lens.
(d) Hypermetropia, corrected with a convex lens.

3. 2002 / 12
Which statement about the retina is correct?

(a) Each retina has about 7 million rods and 120 million cones.
(b) Cones provide black and white vision in bright light.
(c) Cones are found in the greatest concentration in the fovea centralis.
(d) Rods aid in colour vision when the light is dim.

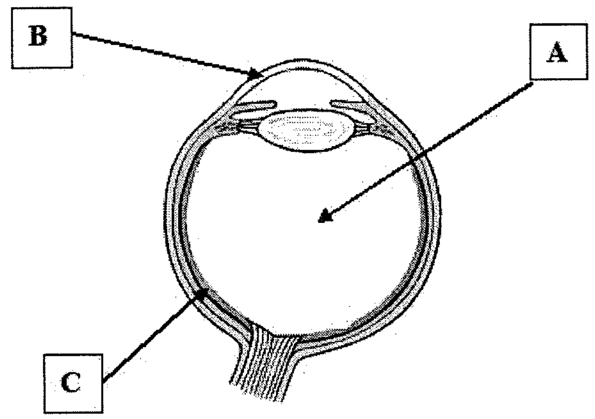
4. 2003 / 28
Which statement about the eye is **CORRECT**?

(a) The sclera separates the eyeball into anterior and posterior cavities.
(b) The lens is held in position by the ciliary body.
(c) Adjustment of muscles in the iris changes the shape of the lens.
(d) Light entering the eye is refracted by both the aqueous and vitreous humour.

5. 2003 / 29
In order to maintain focus on an object as it approaches more closely, the eye will

(a) contract the ciliary muscle causing the lens to bulge.
(b) relax the ciliary muscle causing the lens to bulge.
(c) tighten the suspensory ligaments on the lens and thereby flatten the lens.
(d) dilate the pupil so that more light can be used to accommodate the object.

The next two questions refer to the diagram below.



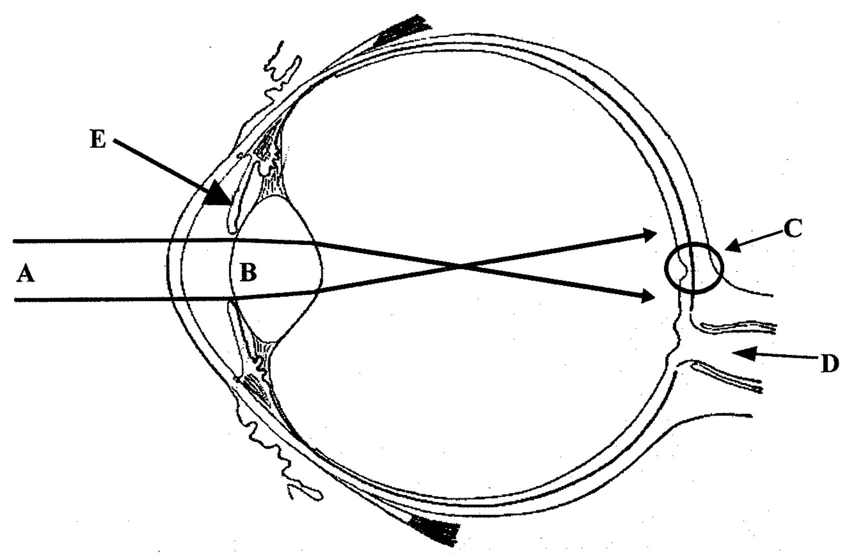
6. 2004 / 17
The structures labelled A, B and C in the diagram above are respectively

(a) aqueous humor, conjunctiva and sclera.
(b) vitreous humor, cornea and retina.
(c) vitreous humor, cornea and sclera.
(d) sclera, conjunctiva and choroid.

7. 2004 / 18
In the condition known as Myopia, an object at a distance to the eye will be focussed

(a) in front of the retina because the eyeball is too long.
(b) behind the retina because the eyeball is too short.
(c) on the retina but will appear smaller.
(d) on the retina but will appear unclear.

8. 2001 / 41
The diagram below shows a human eyeball in longitudinal section.



(2001 / 41 cont)

(a) The arrows starting at A indicate a possible light pathway through the eyeball. What type of sight defect is being illustrated here?

____ (1)

(b) What is structure B?

____ (1)

(c) Apart from B, name **TWO** other components of the eye that bend (refract) the light passing into it.

____

____ (2)

(d) What specific region of the retina is the arrow pointing to at C (circled)?

____ (1)

(e) What type of photoreceptor is present in the highest concentration in C?

____ (1)

(f) What is special about this region of the retina for normal human vision?

____ (1)

(g) Name structure D.

____ (1)

(h) What is the function of structure D?

____ (1)

(i) Name structure E.

____ (1)

(j) Describe the effect that structure E has on the pupil in response to bright and to dark light conditions.

____

____

____

____ (2)

THE EAR

9. 1999 / 16
The part of the ear that detects head position is the

(a) auditory ossicles.
(b) tympanic membrane.
(c) eustachian (auditory) tube.
(d) semicircular canals.

10. 2000 / 17
Sensory hair cells can be found in all of the structures listed below **except** the

(a) auditory canal.
(b) cochlea.
(c) semicircular canals.
(d) vestibule.

11. 2001 / 5
The ossicles are the major structures of the

(a) cochlea.
(b) middle ear.
(c) inner ear.
(d) semicircular canals.

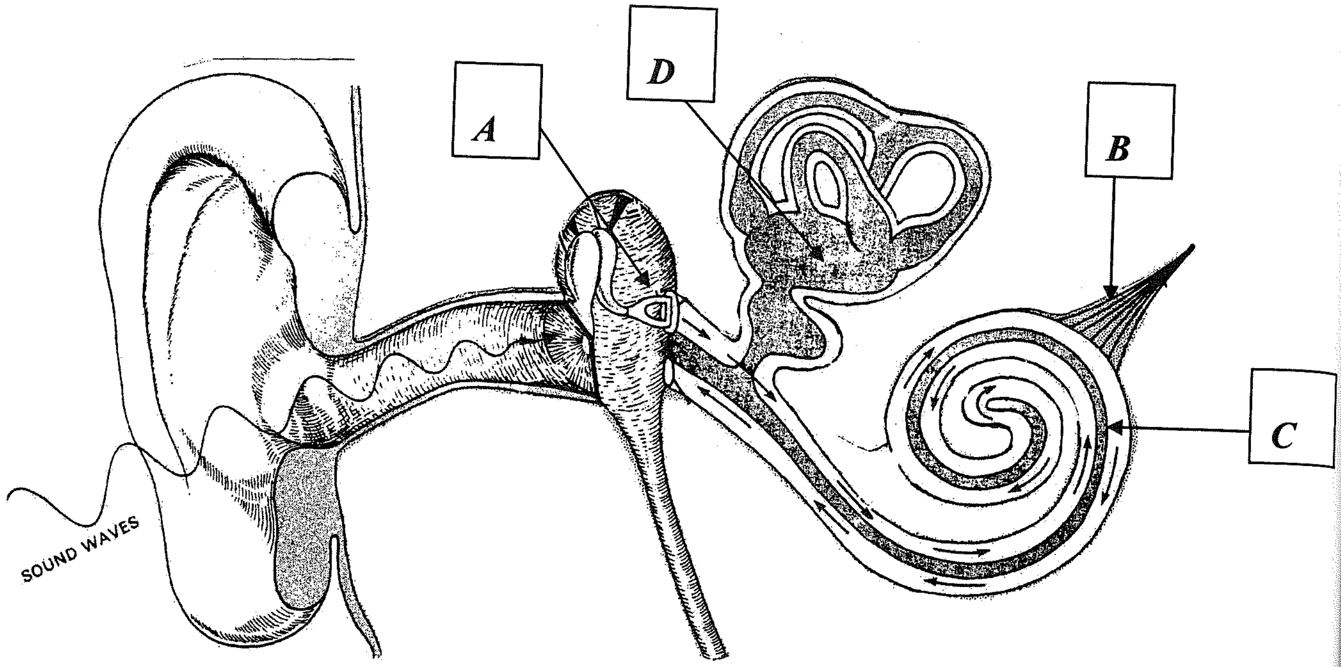
12. 2001 / 8
Receptors for hearing are located in the

(a) tympanic membrane.
(b) semicircular canals.
(c) vestibule.
(d) cochlea.

13. 2002 / 23
A person is able to hear noises of a high frequency but is unable to hear noises of a low frequency. The most likely area of ear damage is the

(a) oval window.
(b) semicircular canals.
(c) ossicles.
(d) cochlea.

The next two questions refer to the diagram below.



14. 2004 / 31
After stepping off an amusement park ride which involves spinning around, a person may experience dizziness because

(a) the otoliths in structure D continue to move around.
(b) the fluid and sensory hairs in structure C are disturbed by the spinning motion.
(c) structure A can be pulled free from the inner ear.
(d) structure B overstimulates the cerebrum.

15. 2004 / 32
Factory workers who have spent many years working next to loud machinery will be able to hear low frequency sounds but will experience some difficulty hearing sounds that are at a high frequency. This is because

(a) the loud noises may have permanently fractured structure A.
(b) some of the sensory hairs in structure C have been damaged.
(c) loud noises can scar the sensory cells that line structure B.
(d) scar tissue in structure D may interfere with its normal function.

16. 2000 / 49
(a) Sensory impulses are physiologically identical regardless of the type of receptor involved. However, they give us completely different sensations. State briefly how this is accomplished.

(1)

(b) Answer this part with reference to the diagram of an ear, below.

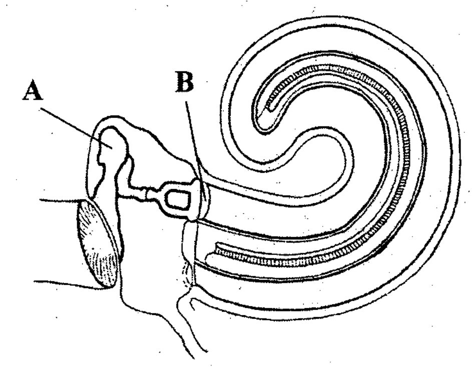


Diagram adapted from Newton and Joyce "Human Perspectives" 2nd Edition 1990.

(i) Describe the function of structure **A**.

(1)

(ii) If structure B were to harden and become rigid, describe the effect this could have on the hearing process.

(1)

EXTENDED ANSWERS

17. 1999 / 51
The body uses many different types of receptors to sense its environment. Through special senses such as vision and hearing, we can detect very complicated information. Describe how receptors in the eye and ear receive and process variable light and sound information, and communicate this to the brain.

(20)

18. 2003 / 50
(a) A major form of deafness is conductive deafness. Define this term and describe its causes.

(5)

(b) Describe how normal balance, posture and locomotion are controlled by the human body.

(15)

4. Nervous System - Neurons

General Objectives

- **Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system**

Content

The neuron as the structural and functional unit of the nervous system – cell body, dendrites, axon, myelin sheath, nodes of Ranvier, motor end plate. Structure and function of receptor, connector and effector neurons.

1. 2000 / 14

In a neuron, the impulse is conducted through the dendrite to the

(a) myelin sheath.
(b) axon.
(c) associated sensory neuron.
(d) cell body.

2. 2002 / 10

Which statement about nerve fibres is correct?

(a) Fibres that have a larger diameter will conduct impulses slower than those with a smaller diameter.
(b) Myelinated fibres conduct impulses slower than unmyelinated nerve fibres.
(c) Nodes of Ranvier increase the speed of nerve impulse transmission.
(d) All myelinated nerve fibres end on skeletal muscle fibres.

3. 2003 / 11

Voluntary muscle movement is controlled by

(a) an upper motor neuron in the brainstem and a lower motor neuron in the spinal cord.
(b) an upper motor neuron in the cerebrum and a lower motor neuron in the dorsal root ganglion.
(c) an upper motor neuron in the cerebrum and a lower motor neuron in the spinal cord.
(d) an upper motor neuron in the spinal cord and a lower motor neuron in skeletal muscle.

The next two questions refer to the diagram below.

4. 2003 / 18

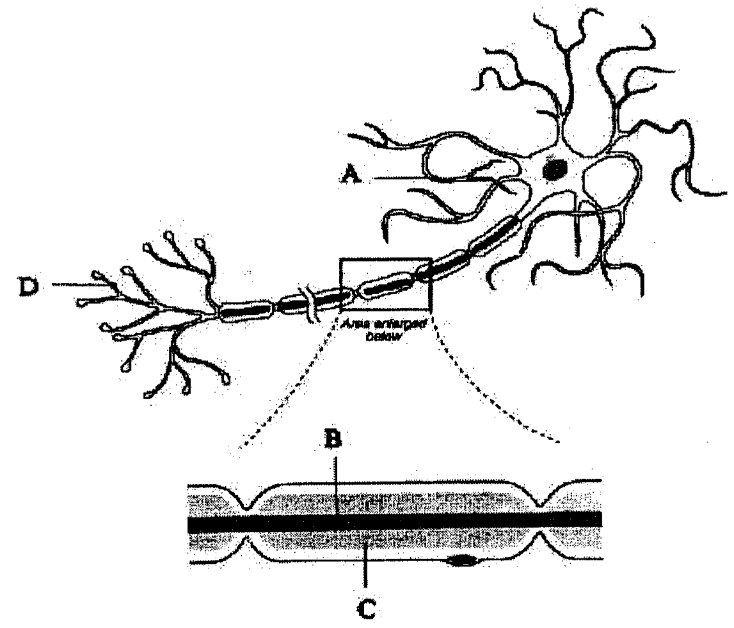
Which part of the structure shown above increases the speed of nerve impulse transmission?

(a) A
(b) B
(c) C
(d) D

5. 2003 / 19

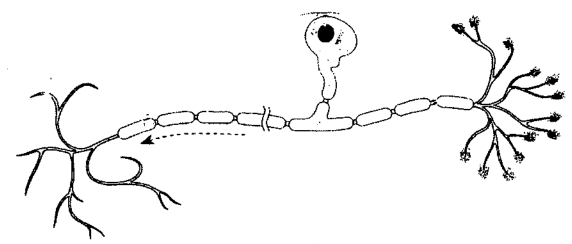
Which one of the following statements about the neuron that follows is correct?

(a) This neuron does not show evidence of a Schwan cell.
(b) Structure B will produce the myelin sheath.
(c) Structure D receives impulses from other neurons.
(d) Structure A transmits impulses to the cell body.



6. 2004 / 6

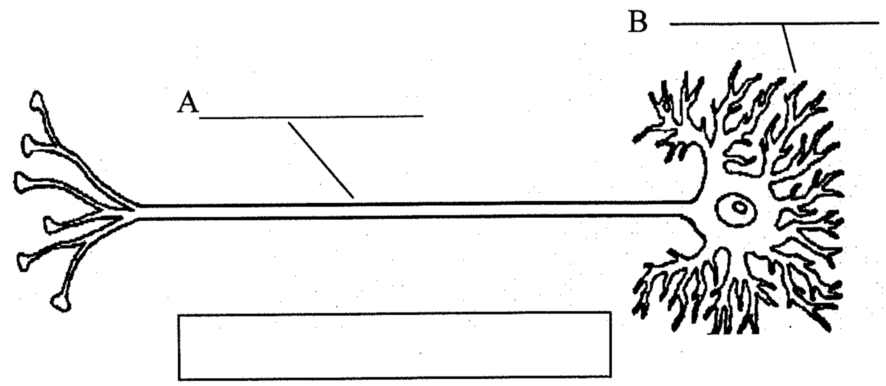
The neuron illustrated below would be classified correctly as



(a) a sensory neuron.
(b) a motor neuron.
(c) a connector neuron.
(d) there is not enough information to say.

7. 2000 / 49

(g) Label the structures A and B of the neuron drawn below. Draw an arrow in the box provided to indicate the direction of the nerve impulse. (3)



DIRECTION OF NERVE IMPULSE

27. Scientific Method

General Aims
To develop competence in the:
- **Process skills of science associated with designing and performing controlled experiments, collecting, recording, presenting and interpreting data**
- **Measurement of length, volume and mass, and in the calculation of magnifications, field diameters and estimations of dimension of structures in light**

General Objectives
- Demonstrate competence in measurement of length, volume and mass, and in calculations of magnifications, field diameters and estimations of dimension of structures in light microscopy.
- Retrieve information from a variety of sources, collate information into succinct reports, communicate accurately and clearly, both orally and in writing
- Analyse and interpret data presented in graphical and tabular form

The next two questions refer to the table below.

Consider the table below showing data recorded for a person viewing a frightening horror movie.

Time (minutes)	1	2	3	4
Cardiac Output (L / minute)	5	7	20	10

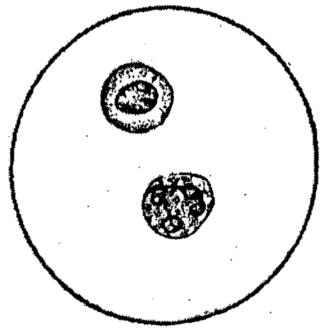
1. 2003 / 26
Concerning these data, which of the statements below is most correct?

(a) The sympathetic nervous system appears to be most active between 2 and 3 minutes.
(b) The parasympathetic nervous system is most active at 3 minutes.
(c) Adrenalin is secreted in response to increased cardiac output at 4 minutes.
(d) Between 2 and 3 minutes the hypothalamus adjusts the heart rate to meet a need for greater oxygen.

2. 2003 / 27
The changes noted in the person's body are

(a) a result of the hormone thyroxine.
(b) a homeostatic response.
(c) designed to maximise oxygen delivery to the skeletal muscles.
(d) designed to ensure that carbon dioxide levels do not increase.

3. 2003 / 32
Cells shown in the diagram below were observed using a microscope fitted with a 40X objective lens and a 10X ocular (eyepiece) lens. Given that the field of view for this microscope at a magnification of 100X is 200 micrometres, what is the approximate size of these cells?



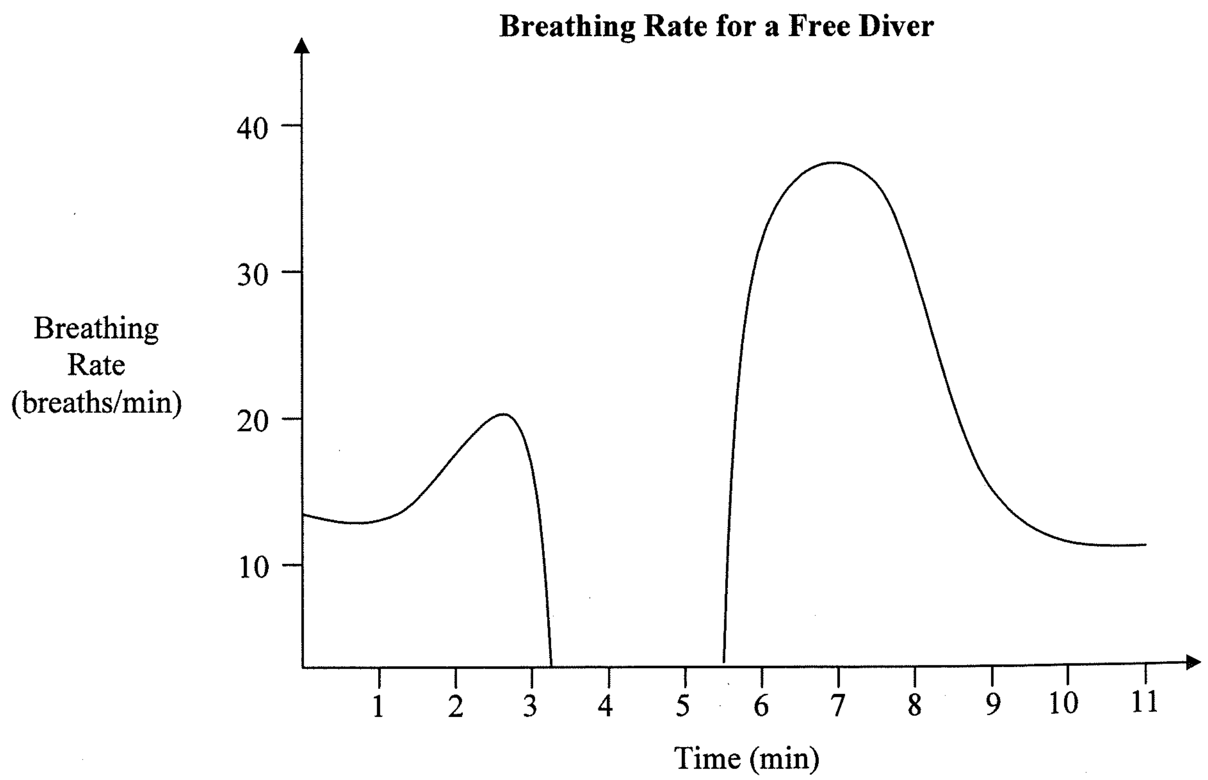
(a) 10 micrometres
(b) 20 micrometres
(c) 30 micrometres
(d) 50 micrometres

4. 2003 / 33
If human cells were given essential nutrients and allowed to grow and reproduce in a closed test tube, what changes would you expect to measure in the test tube?

(a) An increase in temperature, as well as increased levels of oxygen and carbon dioxide.
(b) An increase in temperature, carbon dioxide levels and glucose but decreased oxygen levels.
(c) An increase in temperature and carbon dioxide levels but decreased levels of oxygen and glucose.
(d) A decrease in temperature as well as a decrease in levels of oxygen, carbon dioxide and glucose.

5. 2004 / 25
Free-diving is an Extreme sport in which divers hold their breath for long periods of time and dive to great depths. An investigation into the effect that such a dive has on breathing rate was performed.

The data collected for breathing rate are shown below.

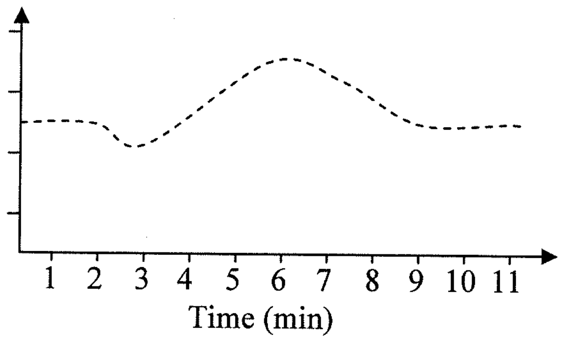


Examine the graphs drawn below. Which of the graphs best represents the change in carbon dioxide levels in the blood during the free dive investigation?

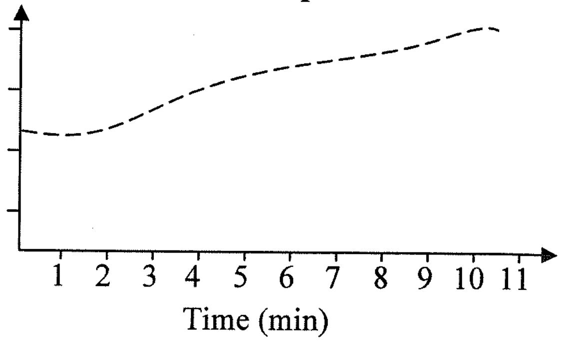
(a) Graph A
(b) Graph B
(c) Graph C
(d) Graph D

Note: The Y-axis represents levels of Carbon Dioxide in the blood.

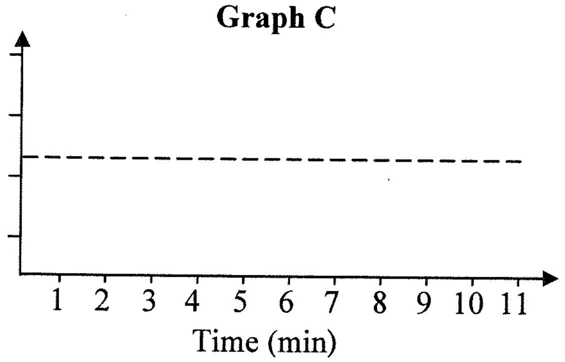
Graph A



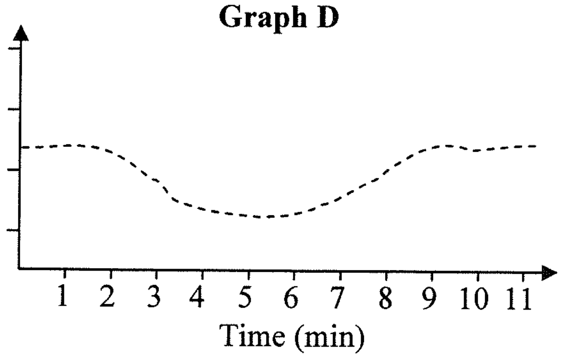
Graph B



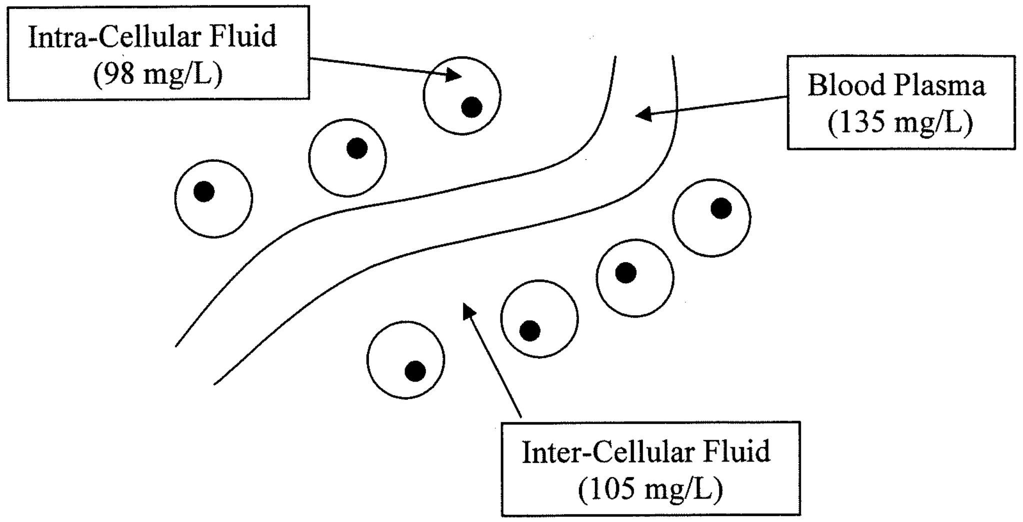
Graph C



Graph D



6. **2004 / 35**
The diagram below indicates part of an image viewed down a microscope. Three different fluid compartments within a small sample of human tissue are shown. Also provided are data on the concentration of total dissolved substances within these compartments.



If you were asked to measure the size of these cells, which piece of information would be most useful?
(a) The field of view.
(b) The magnification of the image.
(c) The number and size of lenses in place.
(d) The type of cells being viewed.

7. **1999 / 45**
A drug company has developed a new influenza vaccine that is squirted into the nasal passages. The vaccine has been approved for testing in people, to see if it prevents the symptoms of influenza in people at risk of infection. Consider how an experiment might be designed to test this vaccine, and answer the following questions.

(a) State a suitable **hypothesis** for this experiment. (1)

____

(b) What would be the **independent variable** in the experiment? (1)

____

(c) What would be the **dependent variable** in the experiment? (1)

____

(d) In this experiment, the subjects would be randomly assigned to two different groups. Using your understanding of scientific method, what name would you use to describe each group, and what would you give to the members of each group to test your hypothesis? (4)

____

____

____

____

(1999 / 45 cont)

(e) List **two** variables that would need to be controlled in this experiment.

(i) ______________________________

(ii) ______________________________ (2)

(f) What sort of immunity is being induced by a vaccine?

______________________________ (1)

(g) Describe **three** barriers present within the nasal cavity that will help to keep infection out of the body.

(i) ______________________________

(ii) ______________________________

(iii) ______________________________ (3)

8. 2000 / 46

(c) You are designing an experiment to test a new drug to be used as a replacement hormone for people with damaged thyroid glands. There are 100 such people in the experiment and you divide them into two groups, a Control group and an Experimental group, to test your drug. Explain how you will choose who goes in which group, and why you have chosen in this way.

____

____

____

______________________________ (2)

9. 2002 / 45

A scientist was investigating the effectiveness of a newly developed drug, "Neurogen", and its ability to improve recovery from nerve cell injury. The experiment was performed as described below:

Nerve cells were placed into culture medium (a solution containing nutrients to keep the cells alive and help them grow). An equal number of nerve cells (in culture medium) were transferred into nine test tubes and maintained under constant temperature and oxygen levels. Three of the test tubes received no further treatment. Of the remaining six test tubes, three tubes received a large, single dose of Neurogen whilst the remaining three tubes had a series of smaller doses of Neurogen added over a one week period.

After one month the growth of nerve axons in the test tubes was measured and the results are shown in the table below.

(2002 / 45 cont)

TREATMENT	Tube number	GROWTH OF NERVE AXONS AFTER ONE MONTH (mm)
NEUROGEN (Large, single dose)	1	0.50
	2	0.42
	3	0.45
NEUROGEN (Series of small doses)	4	0.80
	5	0.98
	6	0.90
NO TREATMENT	7	0.25
	8	0.27
	9	0.19

For the experiment described above:

(a) Identify **ONE** hypothesis being tested. (2)

____

____

(b) What is the Dependent Variable? (1)

____

(c) Based on the information given, list **three** variables that were controlled in this experiment.

(i) ______________________________

(ii) ______________________________

(iii) ______________________________ (3)

(d) What is the purpose of having the three test tubes that did not receive any treatment with Neurogen? (2)

____

____

(e) What **two** major conclusions could be drawn from these results? (2)

(i) ______________________________

(ii) ______________________________

(f) After this experiment, the scientist repeated the procedure three more times. Why would the scientist repeat the experiment so many times? (2)

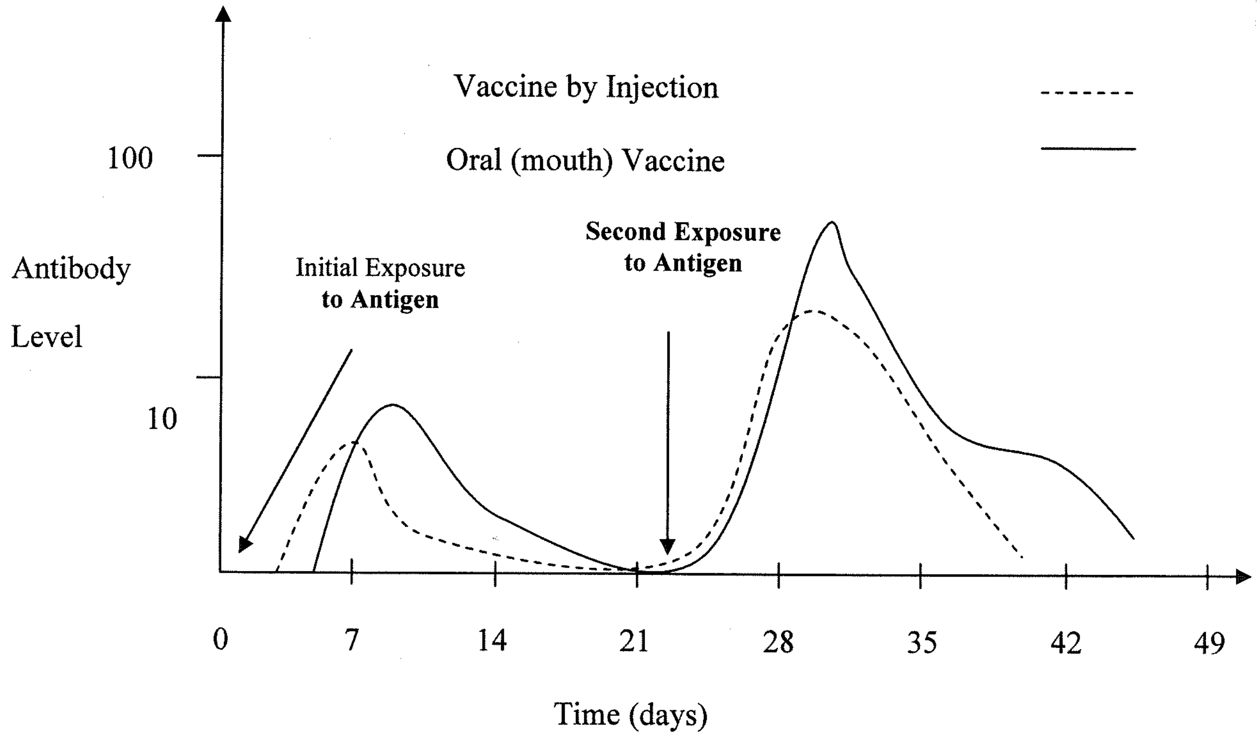
____

____

10. 2003 / 42

In 1968 a new variety of influenza appeared in Hong Kong. Named the " Hong Kong Flu", it began to spread around the world quickly. In many laboratories around the world potential vaccines were investigated and trials performed to assess their effectiveness. The data recorded from one such trial are shown below.

Examine these data and answer the questions that follow:



(a) For this experiment, identify the:

Experimetal Variable : ______

Dependent Variable : ______ (2)

(b) Compare the primary responses shown by the two groups of subjects in the experiment. (2)

(c) Identify **four** factors that would need to be controlled in the selection of subjects for this experiment. (2)

(d) Identify one method by which the vaccines used in the experiment could have been prepared. (1)

(2003 / 42 cont)

(e) Explain clearly, why the secondary response is quicker and larger than the primary response. (3)

(f) Given that an antibody level of 10 is effective at combating the disease, which vaccine would be more successful at controlling the spread of this flu? Explain your answer. (2)

(g) Apart from antibody levels, what other blood measurement could be taken to assess the effectiveness of the secondary responses? (1)

11. 2004 / 43

As the scientist on duty in a regional hospital you receive a sample of Cerebrospinal Fluid (CSF). You are asked to test the sample for a bacterial infection, which you find, and then you are asked to determine the best form of treatment.

As a young scientist you are keen to investigate a drug company's hypothesis that:

"Our new drug 44-Maxophage improves the action of antibiotics."

(a) When designing an experiment to test this hypothesis, what would be

(i) the independent variable?

(ii) the dependent variable?

(iii) two control variables? (4)

In order to test the hypothesis, you grow the bacteria cells in a series of petri-dishes containing essential nutrients. The results of your experiment are shown in the table below.

- A negative (–) indicates that the antibiotic had no effect.
- A positive (+) indicates that the antibiotic destroyed the bacterium.

Note: At higher concentrations, antibiotics can harm the cells of the body.

(2004 / 43 cont)

Antibiotic Concentrate	Petri Dish 1 Antibiotic A only	Petri Dish 2 Antibiotic A and 44–Maxophage	Petri Dish 3 Antibiotic B only	Petri Dish 4 Antibiotic B and 44–Maxophage
100 mg/ml	(+)	(+)	(+)	(+)
50 mg/ml	(+)	(–)	(+)	(+)
25 mg/ml	(–)	(–)	(+)	(–)
10 mg/ml	(–)	(–)	(–)	(–)

(b) Do the results presented in the table support the drug company's hypothesis? Use data from the table to support your answer.

(2)

(c) Of the four treatments, which would you **MOST LIKELY** recommend to treat the bacterial infection? Why?

(2)

(d) What was the purpose of setting up Petri Dishes 1 and 3?

(2)

(e) Identify another hypothesis that these data could be used to evaluate.

(1)

EXTENDED ANSWERS

12. 2001 / 51

(c) A new drug called buprenorphine has been put forward as a possible treatment for heroin addiction but has not been fully evaluated.

Design an experiment to evaluate the effectiveness of buprenorphine as a treatment for heroin addiction. You are limited to using 200 people in your experiment, and their participation is limited to 6 months. Include in the description of your experiment:

the hypothesis being tested;
the dependent and independent variables;
the control and experimental groups;
the controlled variables;
how you could reduce experimental error.

(10)

SOLUTIONS

1 Homeostasis

1	**C**	2	**B**	3	**C**	4	**B**	5	**B**

6. (d) The brain receives a constant / high supply of blood (rich in glucose) / constant supply of nutrients via CSF (1)
   (e) Nerve impulses / nervous system / nervous reflex (1)

7. (a) Physical barrier/tough, hard - Epidermis
   Cooling of the skin - Sweat glands
   Acid mantle - Sweat glands
   Warm/cool skin - (Dilation and constriction of) blood vessels
   Produce melanin to try and protect from UV rays- Melanocytes
   (Limited) antibacterial function - Sebaceous glands
   Waterproof skin - Sebaceous glands
   Keratin that waterproofs the skin - Keratinocytes
   Detection of pain / pressure / temperature etc (can only use once) - Receptors
   Present antigens to lymphocytes - Langerhans cells
   Insulation - Subcutaneous fat / hypodermis
   Vitamin D production - Blood capillaries of dermis
   Inflammatory response - Blood vessels
   (Any of the above - first five mentioned only) (1 mark for function, 1 mark for structure - total 10 marks)

2 Central Nervous System - Structure and Function

1	**C**	2	**C**	3	**D**	4	**C**	5	**B**	6	**B**
7	**A**	8	**D**	9	**D**	10	**B**	11	**A**	12	**B**
13	**D**	14	**D**								

15. A : Pain Receptor
    B : Sensory / Receptor / Afferent neuron
    C : Connector / Relay / Associate / Inter Neuron
    D : Motor / Efferent / Effector nerve
    E : Motor end plate / Neuromuscular junction
    (1 mark each)

16. (c) Medulla contains cardiovascular or respiratory centres/required to sustain life (1)
    Cerebrum less associated with life sustaining functions associated with higher functions (1)

    (f) To remove waste products/carbon dioxide/acids (1)
    replenish necessary nutrients/oxygen/glucose (l)

17. (a) Identify the labelled structures
    A hypothalamus (1)
    B pituitary / hypophysis (1)
    C cerebrum / cerebral cortex / cortex / frontal lobe / gyrus (1)
    D pons / brainstem (1)

    (b) CSF / trabeculae / blood vessels (1)

18. (b) Sensory receptor (1) (in the foot) is stimulated; sensory neuron (1) carries impulse to the spinal cord/CNS (1); via dorsal root ganglion / dorsal root (1) impulse passes to motor neuron (1) via (at least one) synapse / connector neuron / interneuron (1); motor neuron carries impulse out of the spinal cord via the ventral root (1) to effector (1); effector in this case is muscle cells (1) which respond by contracting to move the leg (1). Response occurs at the level of the spinal cord (brain not needed) (1)
    (Diagrams can only be accepted if referred to in the written answer)

19. (b) Role of Receptors:
    Blood pressure (1) monitored by baro/presso receptors (1) in Aorta / Carotid arteries (1) which provide feedback to Cardiovascular centre/medulla / brainstem (1)
    Chemoreceptors in (1) in aortic/carotid bodies and medulla will monitor carbon dioxide / pH of blood (1)
    Oxygen levels monitored by chemoreceptors in aortic/carotid bodies (1)
    Controlled by autonomic nervous system / brainstem (1) (any four points = max. 4 / 3 marks)

Appropriate Response
Parasympathetic/Ach. stimulation will decrease heart activity (1)
Sympathetic/noradrenalin stimulation will increase heart activity (1)
Action on sino-atrial node/pacemaker (1)
As well as atrio-ventricular node and heart muscle (1)
These control heart rate and stroke volume (1)
Controlled by autonomic nervous system / brainstem (1)
(any three points = max. 3 / 4 marks)

3 Nervous System - Receptor Organs

THE EYE

1	B	2	A	3	C	4	D	5	A	6	B
7	A										

8. (a) Short sight / myopia (1)
(b) Lens (1)
(c) Cornea, vitreous humour, aqueous humour (Any two 1 mark each for 2 marks)
(d) Fovea / macula (lutea) / yellow spot (any one of these) (1)
(e) Cones (1)
(f) Area of best focus / point of focus / visual acuity / High concentration of cones / sharpest vision / colour vision (any one) (1)
(g) Optic nerve/ cranial nerve II (1)
(h) Transmission of impulse/electrochemical / sensory message to brain (1)
(i) Iris (1)
(j) In bright light the iris constricts/contracts/makes smaller the pupil (1)
while in dark conditions it dilates/expands/makes bigger the pupil (1)
could describe one then say the other is the opposite. (1 mark each for 2 marks)

THE EAR

9	D	10	A	11	B	12	D	13	D	14	A
15	B										

16. (a) Interpreted in different parts of the brain (1)
(b) (i) Passes vibrations onwards (to incus/stapes) / amplifies / conducts (1)
(ii) (Reduces) vibrations passed on from oval window/reduced nerve impulses generated/cochlea (or other organs) not stimulated
("Becomes deaf" is not accepted as not a process) (1)

17. **(10 marks for eye, and 10 marks for ear - Total 20 marks)**
Eye:
Light travels through to the back of the eye (1) through transparent cornea and fluid to reach the retina (1) which contains rods (1) and cones (1), cells which are stimulated by (1)
Light sensitive pigments break down in response to light (1) which initiates a nerve impulse (1)
That is carried to the brain along the optic nerve (1)
Rods are specialised for dim light (1)
Cones are specialised for responding to colour and bright light (1)
pigments in cones respond to light of differing wavelength (1)
Pupil control amount of light entering the eye (1)
Ear:
Sound waves/air vibration (1) travel along auditory canal (1)
Cause vibration of tympanic membrane (1)
Results in movement of bones in middle ear (1) which moves the oval window (1)
This moves the fluid in the inner ear (1) stimulating hair cells (1) in the Organ of Corti (1) in the cochlear (1). This generates nerve impulses (1) that are carried to the brain along the auditory nerve (1)
Pitch is detected because different areas of the Organ of Corti are stimulated by different frequencies of movement of the fluid (1)
Volume is distinguished because loud noises cause more vibration (1)

18. (a) Definition : conductive deafness due to interference of vibration transmission (1)
Causes : blockage of auditory canal (eg, wax) (1) tympanic membrane damage/rupture (1) ossicle fusion/damage (1) middle ear infection (1) sudden explosive noise (1) diving / pressure changes (1)
(any four points = four marks)
(b) Sensory : the utricle (1) and saccule (1) have sensory hairs (1) that detect the position of otoliths (1) and thus provide information about the stationary position of the head (1).
Semicircular canals (1) detect head movement (1) due to the presence of sensory hairs (1) in the ampulla (1) that detect fluid movement (1).
Nerve impulses from skeletal muscles (1) proprioreceptors / (eg, pressure receptors) (1) are relayed to the cerebellum / midbrain / brainstem (1) and onto sensory cortex of cerebrum (1) visual cues (1)
(max of 12 marks for sensory points)
Motor : unconscious control (eg, muscle tone) - cerebellum / midbrain / brainstem gives feedback to skeletal muscle (1)
Conscious control: motor cortex of cerebrum to skeletal muscles (1) these occur along spinal nerves (1)
Spinal cord reflexes play a role (1) Antagonistic muscles (1) (max 4 marks for motor points)

4 Nervous System - Neurons

1	D	2	C	3	C	4	C	5	D	6	A

7. (g) A = Axon/myelin sheath/neurolemma/nerve fibre (any one : 1 mark)
B = Dendrite/dendritic tree (any one : 1 mark)
Direction = right to left ← (1)

5 Hormones

1	D	2	A	3	D	4	C	5	C	6	C
7	D	8	D	9	A	10	A	11	D	12	C
13	D	14	C	15	B						

16. (a) **A.** Pituitary **B.** Thyroid **C.** Adrenals **D.** Pancreas (4)
(b) (i) Pituitary produces TSH (Thyrotrophin) (1)
(ii) Thyroxine released from the thyroid (1)
(c) (i) Insulin, Glucagon (1)
(ii) Blood glucose concentrations could not be controlled (1)

17. (a) Receptor/binding site (1) on/in target cells specific (1) for only that hormone to bind to.

18. (a) Decreased metabolic rate/obesity or gained weight/lethargy or fatigue/hair loss and pale, thick, dry skin/reduced reproductive efficiency/poor tolerance to cold/decreased body temperature/decreased pulse rate/decreased nervous system function/reduced appetite/growth retardation/cretinism/decreased glucose metabolism/decreased protein synthesis/low blood pressure
Increase in TSH secretion
Goitre/ enlarged thyroid (any one of these for 1 mark)
(b) Increased TSH (1)
No negative feedback (from thyroxine and its effects on TSH release) (1)

19. (a) Oestrogen / oestradiol (1)
(b) Progesterone (1)
(c) Any 3 days between Days 9 & 13 (1)
(d) Ovulation / release of egg / LH surge or release / oestrogen or hormone A drops (1)
(e) Corpus luteum / yellow body (1)
(f) LH / luteinizing hormone (1)
(g) Day 22 (1)
(h) Day 24 or 25 (1)

20. (c) Infundibulum / pituitary or hypophyseal stalk (1)
(d) Will stop releasing/secreting hormones (not producing) (1)
(e) Axons / nerve fibres (1) from hypothalamus (1) are severed or hormones cannot pass to pituitary (1)
(f) Dilute urine / difficulty in uterine contractions during labour (birth) / affect (increase) urine production / affect birth / affect lactation / ADH levels will fall / oxytocin levels will fall / dehydration
(any one : 1 mark)

AIR QUALITY

- Due to fossil fuel combustion (1)
- Increase in particulate matter / photochemical smog (1)
- Increase in sulphur/nitrogen/carbon oxides and sulphides (1)
- Increase in heavy metals such as lead (1)
- Lead to increase in respiratory diseases (1)

OR

- Due to fossil fuel combustion (1)
- Increase in sulphur/nitrogen/carbon oxides and sulphides (1)
- Acidification of atmosphere (1)

(any three points = 3 marks)

- Due to fossil fuel combustion (1)
- Increase in carbon dioxide, methane, nitrous oxides (1)
- Act to increase atmospheric temperatures (1) ie greenhouse effect

OR

- Due to increase CFC levels (1) from foams/plastics/propellants/coolants (1)
- Ozone depletion (1) will occur increasing levels of UV radiation (1)

NOTE: For greenhouse – Max. Mark = 2 out of 3 : For ozone depletion – Max. Mark = 1 out of 3

WATER QUALITY

- Eutrophication (1)
- Excessive nutrient/sewage/fertiliser run-off into water ways (1)
- Increase organic content/algal bloom (1)
- Increase decomposer activity (1)
- Decrease oxygen levels and aquatic life (1)

OR

- Acid rain (1) because of
- Increased levels of sulphur oxides in air from fossil fuel combustion (1)
- Increase acid levels in water systems (1)

OR

- Heavy Metals (1)
- From a variety of industrial/manufacturing processes (1)
- Accumulate in food chains (1)
- Affect humans in a number of ways: nausea, birth defects, nervous disorders etc. (1)

OR

- Increase in salinity of fresh water (1)
- Due to large scale land clearing (1)
- Rising water table increases soil salinity (1)
- and thus run-off into streams (1) (any three points = 3 marks - first example only marked)

27 Scientific Method

1	A	2	D	3	A	4	C	5	A	6	A

Scientific Method questions are often very difficult. The following comments are an attempt to explain why each answer for the multichoice is correct.

1. A Sympathetic system increased heart rate. Other answers not related to information.
2. D Increased activity of heart requires reduced CO_2 as primary response.
3. A Magnification is 40 x 10 = 400X. FOV is 200µm at 100X so is 50µm at 400X. Cells occupy approx 1/5 of FOV circle = 10µm.
4. C All parameters fit the process of respiration for the cells. Others have at least one error.
5. A When diver holds breath the level of CO_2 increases as unable to expel from blood until breathing restarts.
6. A FOV is essential to all measurements as it define the size of the cells within.

7. (a) Administration of the vaccine to people at risk of infection with influenza will prevent symptoms of the disease (accept any valid hypothesis that is a STATEMENT not a question
(must have *variable : cause : variable* relating to the variables in the question)
(if / then format acceptable) (1)
(b) The vaccine (1)
(c) The symptoms of influenza (1)
(d) Treatment/experimental group (1) gets the vaccine (1)
Control group (1) gets a placebo (1)
(e) Age of subjects / other diseases / exposure to same strain of virus / other drugs used / activity level of subjects (anything reasonable) (any two : 1 mark each)
(f) Active artificial (1)
(g) Mucous membrane covered in protective mucus: stops virus sticking to cells (1)
Protective antibodies: prevent virus adhering to cells, damage or kill virus (1)
Cilia: enhance removal of organisms (1)
Passage of air/sneezing: removes infectious organisms in air. (1)
(anything reasonable well described) (one mark each : max 3 marks)

8. (c) Random allocation to each group / stratified random design / even representation (e.g. gender, age, weight etc) in both groups (1)
Accounts for variation between people / reduce bias / reduce effects of chance (1)

9. (a) Neurogen increases / decreases / affects (1) nerve regeneration (1)
OR a series of small doses of Neurogen is more effective than a single dose on nerve regeneration (2).
Note: Must link dependent and independent variables. (If / Then style acceptable of link variables)
(b) Growth / length of nerve axons (1)
(c) Number of cells (1), oxygen level (1), temperature (1), nutrients (1), time (1) culture medium (1) (Any three : 1 mark each)
(d) Control group (1) that provides a comparison for the experimental group (1)
Or eliminates some other factor causing the result (1) (2)
(e) Neurogen promotes growth of nerve axons (1);
A series of Neurogen treatments is more effective than one large dose (1) (2)
(f) Increased reliability / validity / confidence (1);avoid results being due to error / chance (1) (2)

10. (a) Experimental variable : mode of vaccine delivery (1)
Dependent variable : antibody count/level/titre / speed of response (1)
(b) Oral vaccine is slower to cause response (1) or vica verca
Oral vaccine remains at higher level for longer (1) or vica verca
Oral vaccine has a larger response (1) (any two for 2 marks)
(c) age, sex, health, exposure to flu, dosage etc. (any four for 2 marks; three or two is 1 mark)
(d) Deactivated virus / modified virus / weakened virus / dead virus / attenuated virus / viral antigen / genetically-modified microorganism (1)
(e) Vaccine has stimulated production of **memory** cells (1) Antigen recognition quicker (1) Antibody production is quicker (1)
(f) Injection quicker response (1) means that there is **less chance to pass it on** (1)
(g) Level of T-cells or level of B-cells or lymphocytes or WBC count or monocytes or granulocytes or neutrophils or viral antigen (1)

11. (a) (i) 44-Maxophage /drug (1)
(ii) Action of antibiotics or cell death or survival rate (1)
(iii) Antibiotic **OR** type of bacteria **OR** concentrations **OR** Temperature (any two = 2 marks)
If use patient line ie sex, age, fitness (max of 1 mark)
(b) No (1) Petri dishes with 44 Maxophage require higher concentrations of antibiotic (1)
(c) Petri dish 3 i.e. Antibiotic B only (1)
Requires lowest concentration of antibiotic (1)
(d) Act as control/comparison (1) for 44 Maxophage/drug action (1)
(e) Antibiotic concentration increases/decreases survival of bacteria (1)
OR High levels of antibiotic increases/decreases survival of bacteria (1)
OR Antibiotic A is more effective than antibiotic B at killing bacteria (1)
OR Maxophage works better with antibiotic A rather than antibiotic B (1)
(f) Puncture membranes **OR** Puncture cell walls **OR** Inhibit enzymes **OR** Inhibit DNA synthesis
OR Inhibit protein synthesis (any two = 2 marks)

12. (c) A suitable hypothesis for this experiment is that "buprenorphine reduces heroin dependence / addiction"
(accept any valid hypothesis that is a STATEMENT not a question
(must have *variable : cause : variable* relating to the variables in the question)
(if / then format acceptable) (1)
Independent variable : buprenorphine treatment (1)
Dependent variable : heroin dependence / craving for drug (1)
Experimental group must be heroin addicts (1)
Control group receives a placebo (1)
The treatment group receives buprenorphine treatment (1)
(Total 6 marks)
Controlled variables would be : number in group (100) / even group size / dosage / time of day dose is given / age / sex / body weight / diet / length of heroin addiction / no other drugs / etc (Any three : 1 mark each)
Experimental error can be reduced by careful attention to controlled variables / repeating the experiment / increasing sample size etc (anything reasonable) (Any three : 1 mark each)
(Total of controlled and error reduction points : max 4 marks)

Chapter	1999	2000	2001	2002	2003	2004
1.Homeosta	27	**49**	7	8,25,26,**50**		
2.CNS	13,15,29,**49**	15,16,**49**	3,6	11,**42,49**	12,13,17,**49**	1,2,3*,5
3.Receptors	14,16,**51**	17,**49**	5,8,41	9,12,23	28,29,**50**	17,18,31,32
4.Neurons		14,**49**		10	11,18,19	6
5.Hormone	30,**46,52**	23,24,25,**43,46**	1,4,9,17,**42**	4,6,7,**42,49**	1,21,34,**44**	4,**45,47**
6. Drugs	28	26,27,**52**	**51**	**46**	38	
7. Heart	31	28,33,**54**	**43**	28		19
8. Breath	33	**48**	**50**	27	39	**50**
9. Sugar	32	**43**	10	29,30	**41**	**50**
10. Temp	34,35,**47**	34,**54**	2,11,12	**44**	6,7	**44**
11. Kidney	**44**	1,2,3	13,19	1,2,3	14	**42**
12. Fluids	17,18,19,20,24,**44**	7,8,9,10,**42**	14,15,16,18,**44**	5,**43,44**	**45,49**	33,34
13. Infect	21,22,23,25,26,**52**	36,37,38,40,**51,53**	20,21,22,23,**45**	34,35,36,**50**	3,9,10	7,8,38,**49**
14.Genetics	2,4, 42, 43, 50	18,19,21,22,**50**	24,25,26,27,28,34,	15,16	20,24	16,39
15.Pedigree	**43**	**55**	**46**	13,14	**43**	**46**
16.Evol/Mec	1,7,**41,50**		33	17,18,19,**51**	22,37,**46**	40,**51**
17. Evol/Evid		**50,55**		**51**	2,**46**	36
18. Fossils	3,5,10,11	6,**44**	29,31,32	20,21,22	25,**51**	20,21
19.Pong/Hom	**41**	**41**	**47**		4,**48**	9,13,14
20.H Evol/Phys	9,12,**41**	4,39,**41**		24,**41**	5,40	10,11,12
21.H. Evol /Cult	**53**	12,**45,56**	30,**49,52**	**41,51**	**51,52**	22,23,24,**48,52**
22. Race		13	36	**47**	31	
23. Variation	6,8,**41**	5,35	35,**48**	31,33,**47**	23,35,36	37,**51**
24. Settlement		11		32	15	15
25. Pop Dynam	36,37,**48**	20,29,30,**47**	**53**	37,40,**48**	30,**47**	**41**
26. Environs.	38,39,40,**48,53**	31,32,**52,56**	37,38,39,40,**49**	38,39,**48,52**	8,16,**47,52**	26,27,28,29,30
27. Sci Method	**45**	**46**	**51**	**45**	26,27,32,33,**42**	25,35,**43**

2004 3 - this question was incorrect in the TEE and thus not included*